2024-2025 Fall
MAT123 Midterm
14/11/2025

1. Given that

$$f(x) = \begin{cases} e^{-1/x^2}, & \text{if } x \neq 0 \\ \\ 0, & \text{otherwise} \end{cases},$$

find all critical and singular points of $f(x)$. Determining the intervals where $f(x)$ increases and where it decreases, find local extreme values.

2. Determine whether each of the statement below is true or false, providing short explanations if it is true and counterexamples if it is false.

a. Every increasing function has inverse.

b. If a function is continuous on an interval I, then it must be differentiable on I.

c. If $f'(c) = 0$ for some point c in the domain of a function f, then $f(c)$ is either a local minimum or a local maximum value of f.

d. If both $f'(x)$ and $f''(x)$ exist, with $f''(x) \neq 0$ and $f'(x)/f''(x) > 0$ on the open interval (a, b), then $f(x)$ must be either increasing or decreasing on (a, b).

3. Compute the following limits.

a. $\displaystyle\lim_{t\to\pi/2} \frac{|\cos t - \sin t|}{|\cos t| - |\sin t|}$

b. $\displaystyle\lim_{x\to\infty} x^{\sin\left(\frac{1}{x}\right)}$

4. Find the point(s) on the curve $2\tan^{-1}(y/x) = \ln\left(x^2 + y^2\right)$ at which the tangent line is horizontal.

5. A point P in the first quadrant of the Cartesian plane is moving along a circle centered at the origin. As the point P moves, the x-intercept of the tangent to the circle at P moves on the x-axis at the rate of 1 unit per second. At what rate is the point P moving when the tangent line has slope -1?

1. For $x \neq 0$, $f'(x) = \dfrac{2}{x^3} e^{-1/x^2}$ by the chain rule. Also,

$$f'(0) = \lim_{x \to 0} \frac{e^{-1/x^2}}{x} \overset{\text{L'H.}}{=} \lim_{x \to 0} \frac{1/x}{e^{1/x}} = \lim_{t \to \pm\infty} \frac{t}{e^t} \overset{\text{L'H.}}{=} \lim_{t \to \pm\infty} \frac{1}{e^t} = 0.$$

Hence,

$$f'(x) = \begin{cases} \dfrac{2}{x^3} e^{-1/x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}.$$

$$\boxed{\begin{array}{c} \text{The only critical point is } x = 0. \text{ There are no singular points.} \\ f \text{ is increasing on } (-\infty, 0) \text{ and decreasing on } (0, \infty). \\ f(0) = 0 \text{ is the only local (and also absolute) minimum value of } f. \end{array}}$$

2. **a.** $\boxed{\text{TRUE.}}$ For x_1, x_2 in the domain of an increasing function f with $x_1 \neq x_2$, if $x_1 < x_2$ then $f(x_1) < f(x_2)$, and if $x_1 > x_2$ then $f(x_1) > f(x_2)$. Thus, $f(x_1) \neq f(x_2)$, so f is one-to-one and has an inverse.

b. $\boxed{\text{FALSE.}}$ $f(x) = |x|$ is continuous on $(-\infty, \infty)$ but not differentiable at $x = 0$.

c. $\boxed{\text{FALSE.}}$ Consider $f(x) = x^3$, whose critical point is $x = 0$ ($f'(0) = 0$). However, $f(0) = 0$ is neither a local maximum nor a local minimum value.

d. $\boxed{\text{TRUE.}}$ Since $f'(x)/f''(x) > 0$, $f'(x)$ and $f''(x)$ must have the same sign and $f'(x) \neq 0$ on (a, b). By the Intermediate Value Theorem, $f'(x)$ cannot change sign on (a, b) without crossing zero. Thus, either $f'(x) > 0$ for all $x \in (a, b)$ or $f'(x) < 0$ for all $x \in (a, b)$, meaning $f(x)$ is strictly increasing or strictly decreasing on (a, b).

3. **a.** Note that for t close to $\frac{\pi}{2}$, $\cos t - \sin t < 0$, $|\cos t| = \cos t$, and $|\sin t| = \sin t$.

$$\lim_{t \to \frac{\pi}{2}^-} \frac{|\cos t - \sin t|}{|\cos t| - |\sin t|} = \lim_{t \to \frac{\pi}{2}^-} \frac{\sin t - \cos t}{\cos t - \sin t} = -1$$

$$\lim_{t \to \frac{\pi}{2}^+} \frac{|\cos t - \sin t|}{|\cos t| - |\sin t|} = \lim_{t \to \frac{\pi}{2}^+} \frac{\sin t - \cos t}{-\cos t - \sin t} = \frac{1 - 0}{-0 - 1} = -1$$

Since both one-sided limits equal -1, the limit is $\boxed{-1}$.

b. Let $y = x^{\sin(1/x)}$. Then $\ln y = \sin(1/x) \ln x$.

$$\lim_{x \to \infty} \sin\left(\frac{1}{x}\right) \ln x = \lim_{x \to \infty} \frac{\ln x}{\csc(1/x)} \quad \left[\frac{\infty}{\infty}\right]$$

Applying L'Hôpital's Rule:

$$= \lim_{x \to \infty} \frac{1/x}{-\csc(1/x)\cot(1/x)(-1/x^2)} = \lim_{x \to \infty} \tan\left(\frac{1}{x}\right) \frac{\sin(1/x)}{1/x} = 0 \cdot 1 = 0$$

Thus, $\lim\limits_{x \to \infty} x^{\sin(1/x)} = e^0 = \boxed{1}$.

4. Differentiating both sides implicitly with respect to x,

$$2\frac{\frac{d}{dx}(y/x)}{1+(y/x)^2} = \frac{\frac{d}{dx}(x^2+y^2)}{x^2+y^2},$$

$$2\frac{\frac{y'x-y}{x^2}}{\frac{x^2+y^2}{x^2}} = \frac{2x+2yy'}{x^2+y^2}.$$

Simplifying gives $y'x - y = x + yy'$, so $y'(x-y) = x+y$, yielding

$$y' = \frac{x+y}{x-y}.$$

For horizontal tangents, we set $y' = 0$, which implies $y = -x$. Substituting $y = -x$ into the original equation,

$$2\tan^{-1}(-1) = \ln(2x^2) \implies 2\left(-\frac{\pi}{4}\right) = \ln(2x^2) \implies 2x^2 = e^{-\pi/2},$$

$$x = \pm\frac{1}{\sqrt{2}}e^{-\pi/4}.$$

Since $y = -x$, the points are

$$\boxed{\left(\frac{1}{\sqrt{2}}e^{-\pi/4}, -\frac{1}{\sqrt{2}}e^{-\pi/4}\right) \quad \text{and} \quad \left(-\frac{1}{\sqrt{2}}e^{-\pi/4}, \frac{1}{\sqrt{2}}e^{-\pi/4}\right)}.$$

5. Let the circle have radius r, so its equation is $x^2+y^2 = r^2$. A point P in the first quadrant can be parametrized as $P(r\cos\theta, r\sin\theta)$ for $\theta \in (0, \pi/2)$. The slope of the tangent line at P is $m = -\frac{x}{y} = -\cot\theta$. The equation of the tangent line at P is

$$y - r\sin\theta = -\cot\theta(x - r\cos\theta) \qquad [y - y_0 = m(x-x_0)].$$

Setting $y = 0$, the x-intercept is $x_0 = r\sec\theta$. We are given $\left.\frac{dx}{dt}\right|_{x=x_0} = 1$. Differentiating $x = r\sec\theta$ with respect to t,

$$\frac{dx}{dt} = r\sec\theta\tan\theta\frac{d\theta}{dt} = 1 \implies \frac{d\theta}{dt} = \frac{1}{r\sec\theta\tan\theta}.$$

The speed of point P moving along the circle is $v = r\left|\frac{d\theta}{dt}\right| = \frac{1}{\sec\theta\tan\theta} = \frac{\cos^2\theta}{\sin\theta}$.
When the tangent line has slope -1, $-\cot\theta = -1 \implies \theta = \frac{\pi}{4}$. At $\theta = \frac{\pi}{4}$:

$$v = \frac{\cos^2(\pi/4)}{\sin(\pi/4)} = \frac{(1/\sqrt{2})^2}{1/\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}} \text{ units/sec}}.$$